

Document Validation Report

for DiMag



Correspondence percentages: 28%

Number of experiments found: 17

Experiment 1. Formulation stated: "For this purpose, first, we manually collected 1282 raw samples from 12G unique scientific articles (Table S1, Supporting Information) with descriptions of synthesis, materials characterizations, and measurements of magnetic hyperthermia and MRI performances in vitro." **Implementation found:** ['NOT FOUND'] **Missing components:** ['Automated or manual data extraction/curation from 12G scientific articles (1282 raw samples)', 'Parsing pipelines for synthesis, characterization, and performance text into structured records', 'Creation of Table S1 or any code linking raw articles to the dataset', 'Data quality control/validation routines for the collected samples'] **Correspondence:** 5.0%

Experiment 2. Formulation stated: "Average magnetic moment of metals in the core was calculated using magnetic moments of each element with help of stoichiometry to better describe composition of nanoparticles (Equation (S1), Supporting Information)." **Implementation found:** ['NOT FOUND'] **Missing components:** ['Computation of average magnetic moment based on elemental magnetic moments and stoichiometry (Equation S1)', 'Data structure containing elemental magnetic moments', 'Functions or code sections performing stoichiometric weighting of elemental moments', 'Any reference to magnetic moment descriptors within preprocessing or feature engineering steps'] **Correspondence:** 0.0%

Experiment 3. Formulation stated: "Average spin of metal elements on the surface was obtained to describe interactions between the surface elements and proton spins responsible for relaxation properties of the nanoparticle (Equation (S2), Supporting Information)." **Implementation found:** ['analysis.py:top-level ML workflow (reads CSV, predicts log10(r1) relaxivity)', 'ml_model_comparison.py:top-level ML workflow (predicts log10(r2) relaxivity)', 'attached_file.py:top-level ML workflow (predicts log10(r1) relaxivity)', 'sar_model.py:top-level ML workflow (predicts SAR, relaxation proxy)'] **Missing components:** ['Computation of average spin of surface metal elements (Equation S2) – NOT FOUND', 'Any function that derives spin-related descriptors from raw structural data – NOT FOUND', 'Implementation of proton–surface-spin interaction modelling – NOT FOUND', 'Code that links calculated average spin descriptor to relaxation prediction pipeline – NOT FOUND'] **Correspondence:** 25.0%

Experiment 4. Formulation stated: "To describe the role of organic coating, we employed a partition coefficient LogP and number of H acceptors, which were calculated using SMILES representations of organic molecules." **Implementation found:** ['NOT IMPLEMENTED'] **Missing components:** ['Computation of LogP from SMILES strings (e.g., with RDKit)', 'Calculation of hydrogen bond acceptor count from SMILES', 'Any code that parses or reads SMILES representations', 'Incorporation of the calculated organic-coating descriptors into the ML workflow'] **Correspondence:** 5.0%

Experiment 5. Formulation stated: "Volume and surface area were calculated using the shape and size of the nanoparticle to describe morphological characteristics of nanomagnets."

Implementation found: ['NOT FOUND'] **Missing components:** ['Computation of nanoparticle volume based on shape and size parameters', 'Computation of nanoparticle surface area based on shape and size parameters', 'Integration of computed morphological descriptors (volume, surface area) into the dataset/pre-processing pipeline'] **Correspondence:** 5.0%

Experiment 6. Formulation stated: "To cope with that, we first removed the total of 211 samples such that each sample in the remaining dataset had no more than 15% of missing values."

Implementation found: ['NOT IMPLEMENTED'] **Missing components:** ['Logic to calculate percentage of missing values per sample', 'Removal of samples with >15% missing values (211 samples) before further processing'] **Correspondence:** 0.0%

Experiment 7. Formulation stated: "After that, we applied the kNN imputation algorithm to fill the gaps and obtain the complete dataset." **Implementation found:** ['NOT FOUND'] **Missing components:** ['kNN imputation step for handling missing values (no use of sklearn.impute.KNNImputer or equivalent found)'] **Correspondence:** 5.0%

Experiment 8. Formulation stated: "Further, to alleviate redundancy in data and retain only the most informative features, we transformed highly correlated geometrical features into area to volume and max to min length ratios reflecting size and shape anisotropy of nanoparticles."

Implementation found: ['NOT IMPLEMENTED'] **Missing components:** ['Generation of area-to-volume ratio feature', 'Generation of max-to-min length ratio feature', 'Identification of highly correlated geometrical descriptors', 'Replacement/removal of original correlated geometrical features after ratio transformation', 'Any code comments or functions explaining size and shape anisotropy feature engineering'] **Correspondence:** 10.0%

Experiment 9. Formulation stated: "We also discarded noninformative features of low variance, i.e., coercivity and remanent magnetization values for prediction of MRI relaxivities."

Implementation found: ['NOT FOUND'] **Missing components:** ['Explicit removal of low-variance features (coercivity and remanent magnetization) prior to model training'] **Correspondence:** 10.0%

Experiment 10. Formulation stated: "Finally, we removed samples with outliers by setting empirical thresholds on feature values while analyzing their distributions." **Implementation found:** ['NOT FOUND'] **Missing components:** ['Outlier removal based on empirical thresholds of feature distributions'] **Correspondence:** 0.0%

Experiment 11. Formulation stated: "After the final dataset has been prepared, we divided it into three, one for each regression problem: prediction of SAR value, r1 or r2 relaxivity containing 4G0, 355, and 4G5 samples, respectively." **Implementation found:** ['script.py (SAR.csv workflow):', 'analysis.py:', 'ml_model_comparison.py:', 'attached_file.py:', 'sar_model.py:'] **Missing components:** ['Single script/function that takes a unified dataset and programmatically splits it into the three target-specific subsets (SAR, r1, r2)', 'Explicit check/enforcement of the reported sample counts (4G0, 355, 4G5) after the split', 'Central experiment driver that runs all three regression problems in one reproducible pipeline'] **Correspondence:** 85.0%

Experiment 12. Formulation stated: "We normalized and log-scaled the data for each problem before splitting to train and test sets in the 80/20 ratio." **Implementation found:** ['analysis.py:main (top-level procedural code, lines ~1-250) – scales features & log-transforms target prior to an 80/20 stratified split', 'script.py (SAR.csv workflow):main – performs feature scaling & log10 target transform, then applies an 80/20 train/test split'] **Missing components:** ['Uniform preprocessing order across all scripts (many other scripts apply scaling/log transform AFTER the split)', 'Explicit guarantee that the split ratio is exactly 0.8/0.2 in every script (parameter not shown in several files)', 'Log-scaling of input features (only the target is log-scaled in all found implementations)']

Correspondence: 55.00000000000001%

Experiment 13. Formulation stated: "In this work, we opted for tree-based algorithms such as Random Forest and Gradient Boosting, as they remain state-of-the-art for midsize inhomogeneous tabular data, on par with or even outperforming modern deep learning models." **Implementation found:** ['script.py:', 'analysis.py:', 'ml_model_comparison.py:', 'attached_file.py:', 'unnamed_script.py:', 'sar_model.py:'] **Missing components:** None **Correspondence:** 95.0%

Experiment 14. Formulation stated: "To achieve the highest model performance for each regression task, hyperparameter tuning was done using the model-agnostic Optuna framework (Table S5, Supporting Information)." **Implementation found:** ['NOT FOUND'] **Missing components:** ['Optuna study definition and optimization loop for hyperparameter tuning', 'Objective function integrating model training and cross-validation within Optuna', 'Recording or saving of best hyperparameters per regression task', 'Integration of tuned hyperparameters into final model training'] **Correspondence:** 5.0%

Experiment 15. Formulation stated: "For each regression problem, we optimized and compared performance of four models (Figure S4, Supporting Information)." **Implementation found:** ['script.py:global (SAR.csv workflow with 4 models)', 'analysis.py:global (r1 relaxivity workflow with 4 models)', 'ml_model_comparison.py:global (r2 relaxivity workflow with 4 models)'] **Missing components:** ['Explicit hyper-parameter optimisation procedure for each model (only fixed parameters present)', 'Central routine that aggregates/compares results across the different regression problems', 'Automated generation of Figure S4 or combined visual summary of all model comparisons'] **Correspondence:** 80.0%

Experiment 16. Formulation stated: "We plotted SHAP (SHapley Additive exPlanations) values for each selected ML model (Figure 3B)." **Implementation found:** ['NOT IMPLEMENTED'] **Missing components:** ['Computation of SHAP values for ExtraTrees, RandomForest, LightGBM, XGBoost models (no shap.TreeExplainer or similar found)', 'Visualization of SHAP importance plots (e.g., summary_plot, bar_plot) for Figure 3B'] **Correspondence:** 25.0%

Experiment 17. Formulation stated: "As an additional validation effort, we formed another testing set of samples (Tables SG and S7, Supporting Information) that have not been used for training and optimizing hyperparameters of the ML models." **Implementation found:** ['script.py (r1

relaxivity, training+external validation CSVs):global scope', 'script.py (r2 relaxivity, training+external validation CSVs):global scope', 'sar_model.py:global scope (held-out/validation set evaluation)']
Missing components: ['Loading of specific Tables SG and S7 sample files (not referenced in any script)', 'Programmatic guarantee that external validation set was never used during hyper-parameter optimisation', 'Evaluation of all four ensemble models on the additional validation set (currently only one or two models are validated externally)'] **Correspondence:** 70.0%